

Project 2: Exploratory Data Analysis (EDA) & Dashboarding

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Track: Data Analytics Internship

1. Project Overview

Building upon the data cleaning phase completed in Project 1, this second phase focuses on Exploratory Data Analysis (EDA). The objective is to analyze the sanitized dataset to uncover sales trends, customer purchasing behavior, and product performance using Microsoft Excel.

2. Descriptive Statistics & Anomaly Detection

- **Summary Statistics:** Calculated key metrics (Mean, Median, Standard Deviation) for quantitative variables to understand the data distribution.
- **Outlier Detection:** Utilized a **Box & Whisker Plot** to analyze the Total Price column. The visualization successfully identified high-value outliers, which likely represent bulk B2B purchases or VIP customer transactions.

3. Correlation Analysis

Conducted a Pearson correlation analysis between Unit Price and Quantity.

- **Result:** The correlation coefficient is approximately **0.028**.
- **Conclusion:** There is **no significant relationship** between the item's price and the quantity purchased. Customers buy the quantities they need regardless of price variations. A **Scatter Plot** was generated to visually confirm this lack of correlation.

4. Interactive Visualizations & Dashboard

An interactive Excel dashboard was developed using Pivot Tables and Pivot Charts to visualize categorical and temporal data.

Key Components:

- **Total Sales Trend Over Time (Line Chart):** Highlights the overall trajectory of revenue across the years 2023, 2024, and 2025.
- **Total Sales by Product (Column Chart):** Compares the volume of sales across different item categories.
- **Sales Distribution by Payment Method (Pie Chart):** Illustrates customer preferences regarding payment channels.
- **Interactivity:** Integrated **Slicers** for Product and Payment Method, allowing real-time, dynamic filtering across all connected charts.

5. Business Insights & Recommendations

- **Product Strategy:** Printers and Chairs are the top-performing products. Marketing efforts and inventory management should prioritize these high-demand items.

- **Sales Decline:** The line chart indicates a drop in total sales in the most recent periods. A deeper investigation into external market factors or internal marketing strategies is recommended.
- **Payment Ecosystem:** The highly balanced distribution across Cash, Credit Cards, Gift Cards, and Online transfers indicates a healthy payment ecosystem. No single method requires targeted optimization.